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High-Power GaN-Based Blue Laser Diodes Degradation Investigation and Anti-aging Solution

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Gallium nitride (GaN)-based semiconductor laser diodes (LDs) have garnered significant attention due to their promising applications. However, high-power LDs face serious degradation issues that limit their practical use. This study investigates the degradation factors of 437 nm and 6.3 W LDs by comparing light–current–voltage (L–I–V) characteristics, transmission electron microscopy (TEM), cathodoluminescence (CL), and secondary ion mass spectroscopy (SIMS) before and after 1000-h aging. The diffusion of mirror coating from the resonant cavity surface is identified as a key factor contributing to high-power LD degradation, which has not been reported in milliwatt-level LDs. Meanwhile, the mechanisms behind the LD degradation are profiled and summarized together with the diffusion and other factors. On basis of the mechanism exploration, an anti-aging technology for high-power GaN-based LDs is developed by using aluminum nitride for passivation layer and sapphire materials for mirror film. This anti-aging technology has been verified, and a nearly ten-time degradation suppression is achieved from 1000 h. This study elucidates the degradation mechanisms of high-power GaN LDs and provides an effective technology to extend their lifespan, thereby prompting the practical applications of high-power LDs.

1. Introduction

Gallium nitride (GaN)-based semiconductor laser diodes (LDs) are widely used in wireless optical communications, industrial manufacturing, and display.[1] With increasing practical application requirements, the demand for higher-power LDs is also rising. In recent years, high-power blue LDs have been continuously developed and fabricated by the scientific and industrial communities. For example, Strauss et al. fabricated a blue LD with an output power of 3.5 W in 2017,[2] and Liang et al. proposed a 442 nm blue LD that reached 6.0 W in 2021.[3] Additionally, our research group developed an 8.0 W LD with a 445 nm wavelength,[4] and the recent record was achieved by Masaya et al. in 2022 with an output power of 13.2 W.[5] Nevertheless, the investigation of degradation mechanisms has mostly focused on milliwatt-level devices. Previous literature has reported the main degradation factors of milliwatt-level GaN-based LDs, including degradation of epilayer, resonant cavity, electrode, and welding.[6] Among these factors, the increased defects in the active region and degradation of the cavity surface could be the main ones. These defects are induced by high current and temperature stresses,[7] which increase structural defects and enhance the diffusion of magnesium (Mg) from the *p*-type layer, resulting in nonradiative recombination centers (NRCs) in the active region.[8] The degradation of the cavity surface can be induced by higher-power density, leading to oxidation of the cavity surface, formation of carbon deposition, and catastrophic optical damage (COD).[6a,6e,6g,6h,9] Anti-aging technologies for milliwatt-level LDs have been developed to improve their lifespan and reliability. For instance, epitaxy techniques such as plasma-assisted molecular beam epitaxy (PAMBE) can minimize defects in the material.[10] Passivation techniques can also be employed to prevent the effects of surface recombination and oxidation, thereby enhancing cavity surface reliability.[11] However, whether these anti-aging technologies are applicable to high-power GaN-based LDs remains uncertain. The higher power of these devices leads to more severe degradation problems, negatively impacting their lifespan and reliability. Therefore, studying the degradation mechanisms of high-power (watt-level) GaN-based LDs is essential for developing new anti-aging technologies and advancing higher power and efficiency in LDs.

In this study, 437 nm and 6.3 W LD was fabricated and conducted to an aging test. The sample before and after aging was then compared to analyze the degradation mechanisms of high-power LDs. The comparisons utilized comprehensive characterization techniques, including light–current–voltage (L–I–V) characterization, transmission electron microscopy (TEM), cathodoluminescence (CL), and secondary ion mass spectroscopy

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DOI: 10.1002/adpr.202400119

(SIMS). Additionally, the influences of elemental diffusion and temperature variations on the L–I–V characteristics of the LD were numerically simulated by using a three-dimensional photonic integrated circuit simulator (PICS3D). Finally, an anti-aging technology for high-power GaN-based LDs was developed by designing and fabricating a new structure. This structure involved introducing an aluminum nitride (AlN) passivation layer to suppress the diffusion of coatings for the mirror and replacing the mirror film to improve its reliability.

2. Results and Discussions

A GaN-based blue LD was epitaxially grown on a free-standing GaN substrate with an asymmetric multiple quantum well (MQW) structure. The blue LD was fabricated based on the ridge-waveguide edge-emitting structure, as illustrated in **Figure 1**. The aging test was conducted under continuous-wave (CW) operation at 1.5 ampere (A) for over 1000 h (h) to characterize the LD degradation.

The light output power is a crucial parameter for evaluating LD performance. The maximum light output power of the LD was initially measured. The light output power reached 6.3 W at a forward current of 5 A (**Figure 2**a). LD degradation typically manifests as a decrease in light output power at a constant drive current.[12] Thus, the light output power of the 6.3 W LD under a 1.5 A injection current was measured at different aging times (Figure 2b). The light output power showed accelerated degradation after 96 h of aging, eventually decreasing to 79.2% of the initial value after 1000 h. The L–I–V characteristics before and after aging were illustrated in Figure 2c, which indicated an increase in the threshold current from 0.463 to 0.629 A and a decrease in the slope efficiency from 1.48 to 1.36 W A^{-1}. These changes suggested a decrease in the energy conversion efficiency of the aged sample and an increase in nonradiative recombination within the active region.[13] Additionally, the I–V characteristics before and after aging showed a slight voltage change (i.e., the series resistance did not dramatically change). To observe subtle changes in the I–V characteristics, a refined measurement was conducted and plotted on a logarithmic scale (Figure 2d). Notably, there was a variation in the forward current at voltages below the turn-on voltage after aging, which could be attributed to the degradation of the *p*–*n* junction.[14] The degradation of L–I–V characteristics had been observed, but further investigation was required to fully understand the mechanisms behind it.

The mechanisms behind LD degradation are complex and can be analyzed through changes in light output power, which typically decreases during operation. This behavior can be expressed by the following equation[15]

$$P_{\text{out}} = \eta_i \left(\frac{\alpha_m}{\alpha_i + \alpha_m}\right) \frac{hc}{q\lambda} \left[I - I' \exp\left(\frac{\alpha_i + \alpha_m}{g\Gamma}\right)\right] \tag{1}$$

where η_i is the injection efficiency, I' is the transparency current, g is the material gain, Γ is the optical confinement factor, and α_i and α_m are the internal and mirror losses, respectively, which constitute the total optical loss of the LD.[16] The decrease in light output power can be attributed to changes in injection efficiency and optical loss of the LD, as indicated by Equation (1). Further analysis can be conducted by decomposing this expression into parameters such as threshold current and slope efficiency, represented by Equation (2) and (3), respectively[15]

$$I_{\text{th}} = I' \exp\left(\frac{\alpha_i + \alpha_m}{g\Gamma}\right) \tag{2}$$

$$\mu_{\text{slope}} = \eta_i \left(\frac{\alpha_m}{\alpha_i + \alpha_m}\right) \frac{hc}{q\lambda} \tag{3}$$

Experimental measurements revealed an increased threshold current in the aged sample, indicating an increase in optical loss during LD operation, as described by Equation (2). To assess this increase, the threshold currents before and after aging were applied to this equation. Based on numerical simulations, the values of I', g, and Γ were set to 300 mA, 1430 cm^{-1}, and 5.8%, respectively. This calculation showed that the optical loss after aging increased by 64% compared to that before aging. In addition to the threshold current, the decrease in slope efficiency also indicated the LD degradation. The slope efficiency before aging was divided by that of after aging (the optical loss values were incorporated into this calculation) and resulted in a ratio of ≈0.66. This ratio could also be represented as

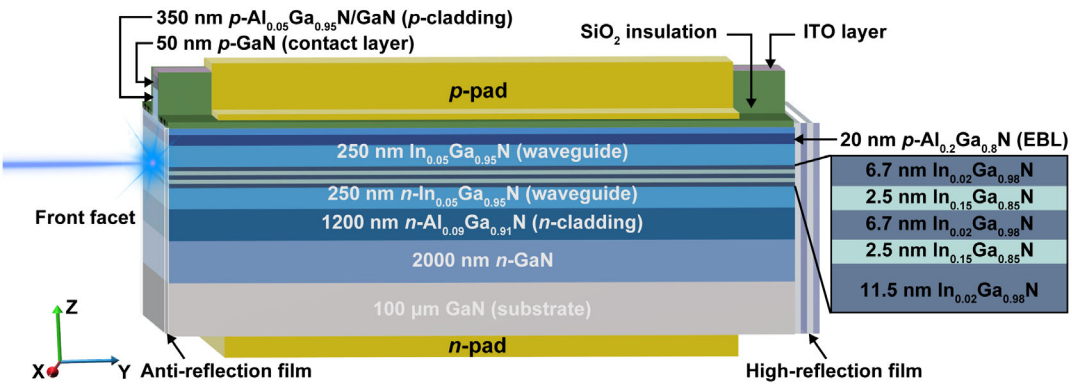

Figure 1. The structure diagram of the ridge-waveguide edge-emitting LD. The ridge-waveguide structure was 30 × 1200 μm^2. The *p*-pad and *n*-pad were deposited on each side for the current injection. To form the *p*-type contact, an indium tin oxide (ITO) layer was deposited. The electron-blocking layer (EBL) was grown between the *p*-cladding layer and the waveguide layer.

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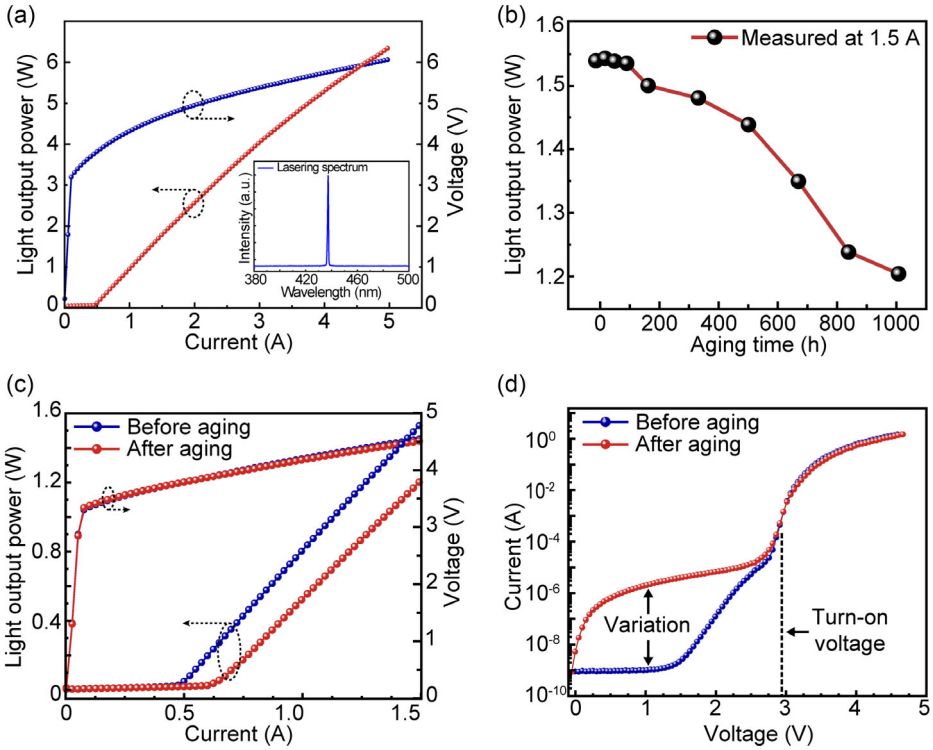

Figure 2. a) Light–current–voltage (L–I–V) characteristics of the blue LD. The sub-figure inset was the lasing spectrum measured at 1.5 ampere (A) and under room temperature. b) The light output power of the LD under continuous-wave operation at 1.5 A at different aging times. c) L–I–V characteristics of the blue LD before and after aging. d) The variation in the I–V characteristics at voltages below the turn-on voltage.

$(\eta_{\text{i_before}} \times \alpha_{\text{m_before}})/(\eta_{\text{i_after}} \times \alpha_{\text{m_after}})$. The value of 0.66 indicated that the deterioration of the carrier injection and the increase in mirror loss were vital contributions to LD degradation. The carrier injection deterioration could be closely associated with changes in carrier distribution within the *p*–*n* junction.[17] Meanwhile, mirror loss is primarily determined by the flatness and reflectivity of mirror films, which increases with the degradation of the cavity surface.[18] Additionally, optical loss could stem from internal loss, involving carrier capture and photon absorption induced by defects and impurities in the active region of the LD epilayer. This internal loss can be exacerbated by defect generation and impurity diffusion during LD operation.[19] These factors collectively reduce LD efficiency, that is, converting more electrical energy into heat and thus increasing the operating temperature, which further accelerates the degradation process.[20] Therefore, the degradation process involving multiple factors requires further investigation.

2.1. Analyses of Degradation Mechanisms

According to Equation (1), LD degradation can be attributed to factors such as optical loss and carrier injection deterioration. Optical loss consists of internal and mirror losses, which are associated with the degradation of the epilayer and cavity surface of the LD, respectively. More importantly, temperature-related factors must be considered.

2.1.1. Optical Loss Caused by Degradation in LD Epilayer

Optical loss in LDs involves internal loss within the epilayer, primarily attributed to structural defects in the active region, such as dislocations and point defects.[21] Previous studies utilizing TEM have observed dislocations near the MQWs of LDs, which significantly diminish the luminescence efficiency.[6b] Therefore, structural and optical characterizations of the LD epilayer were conducted, comparing the ridge and non-ridge regions of aged samples. The ridge region underwent current injection, but the non-ridge region did not (**Figure 3**a). The cross-sectional TEM view of the analysis area was depicted in Figure 3b, and the details of MQW structures were also illustrated. Both the ridge and non-ridge regions showed sharp interfaces in the MQWs, indicating that stress did not induce structural damage to the MQWs. Moreover, no discernible dislocations were observed near the MQWs, suggesting that LD degradation was not caused by dislocations multiplying in the active region. After ruling out the influence of dislocations, the internal loss in the active region might stem from invisible structural defects, such as point defects.

As the point defects in the active region could act as highly effective NRCs, significantly reducing the internal quantum efficiency (IQE) of the LD.[22] To further investigate the degradation of IQE, the optical properties near the active region were characterized using the CL technique. The monochromatic CL image taken at 460 nm of the light-emission region (including the active region), was presented in Figure 3c. To quantify and analyze

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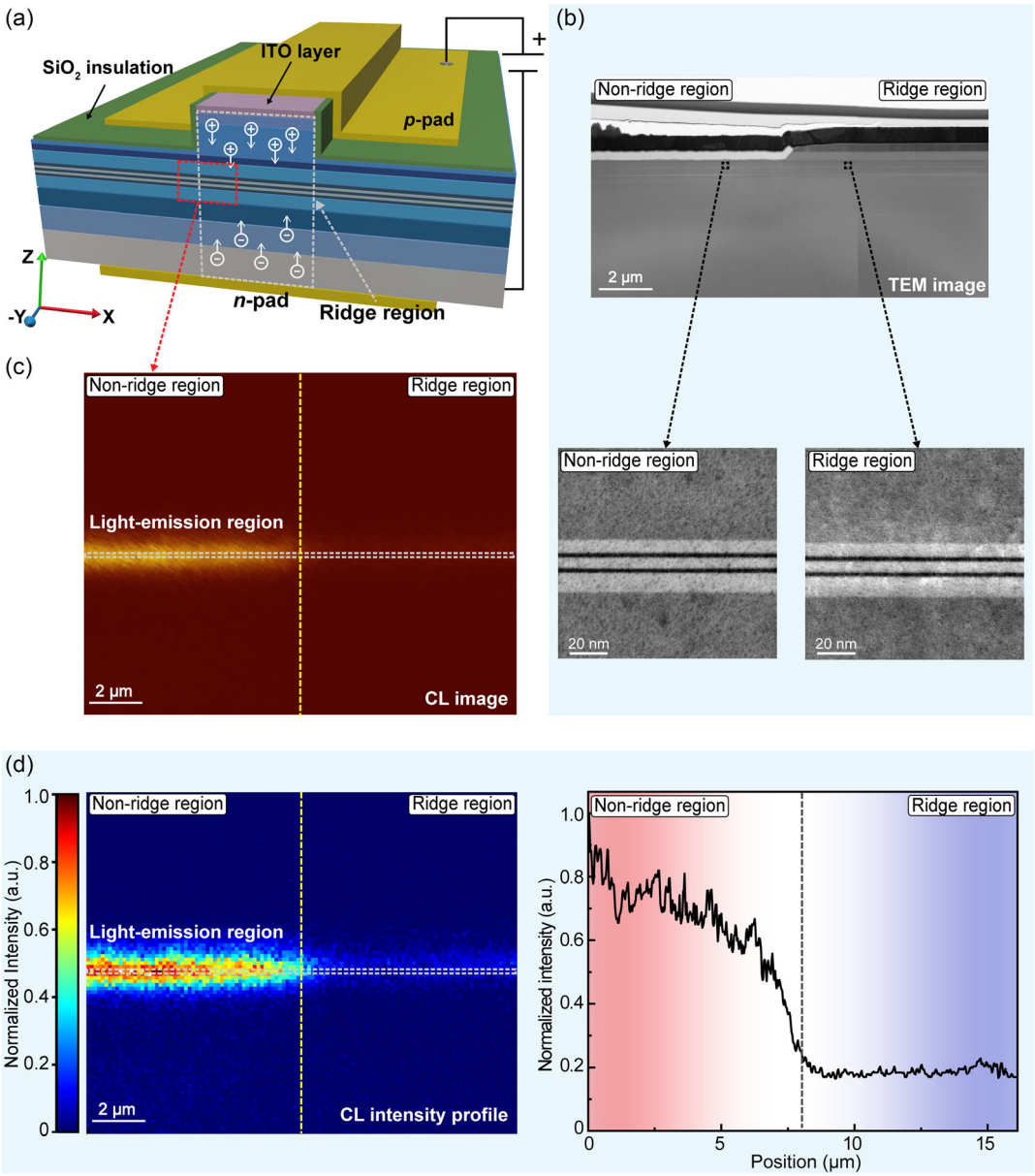

Figure 3. a) The structure diagram of the LD front facet. The anti-reflection film was omitted for a clear presentation of the structure. The area delineated by the dashed line represented the ridge region, which underwent current injection. The rest was the non-ridge area that did not receive the current injection. b) TEM observation of the LD device. The upper sub-figure was the cross-sectional TEM view of the ridge and non-ridge regions. The lower sub-figures were the TEM images of multiple quantum wells in the ridge and non-ridge regions, respectively. c) CL monochromatic (taken at 460 nm) image near the active region. d) Normalized intensity of CL signal. The left sub-figure was the heatmap of the CL intensity profile in the ridge and non-ridge regions. The right sub-figure was the CL intensity profile along the light-emission region.

changes in the optical properties of the light-emission region, the normalized CL intensity profiles were shown in Figure 3d. This analysis revealed a significant decrease in CL intensity in the ridge region, reaching only ≈26% of that in the non-ridge region. The decreased CL intensity also suggested a reduction in the IQE of the ridge region. IQE represents the relationship between the radiative recombination and the total recombination (including nonradiative, radiative, and Auger recombination).[23] It was because the radiative and Auger recombination did not change significantly after aging,[24] the decrease in IQE of the ridge region could be attributed to an increase in the nonradiative recombination (due to point defects).

Point defects could originate from various sources. On one hand, increased point defects in the active region include native defects such as gallium or nitrogen vacancies (V_{Ga}, V_N), gallium or nitrogen interstitials (Ga_i, N_i), and gallium or nitrogen in substitutional positions (Ga_N, N_{Ga}). These defects cannot be characterized by SIMS, and their diffusion processes are

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challenging to evaluate because they involve native atoms. However, V_{Ga} and V_N might form more readily in *n*-type and *p*-type GaN on formation energy calculations.[25] These native defects may propagate or be generated during stress processes, owing to mechanisms like generation or percolation.[8a,26] On the other hand, increased point defects are related to the diffusion of impurities such as Mg and silicon (Si), which may diffuse from the *p*/*n*-type layers into the active region as point defects during LD operation.[27] However, reports on impurity diffusion in GaN-based LDs are limited due to the difficulty in analyzing the narrow area of LDs.[28] Accurate and reliable measurements of impurity diffusion in the LD epilayer are therefore necessary.

The distribution of Mg and Si within the LD before and after aging was further investigated by analyzing their profiles via SIMS (**Figure 4**a). The SIMS results highlighted specific regions within the LD epilayer related to light emission, including the active region and waveguide layers, as illustrated in Figure 4b,c. It was observed that Mg diffused into the active region during operation, and the quantum barrier of the active region showed a high concentration of Si doping. This indicated that the inter-diffusion and the subsequent accumulation of impurities in the active region might contribute to LD degradation. The impurities diffused into the active region, acting as point defects, would result in optical absorption and nonradiative recombination,

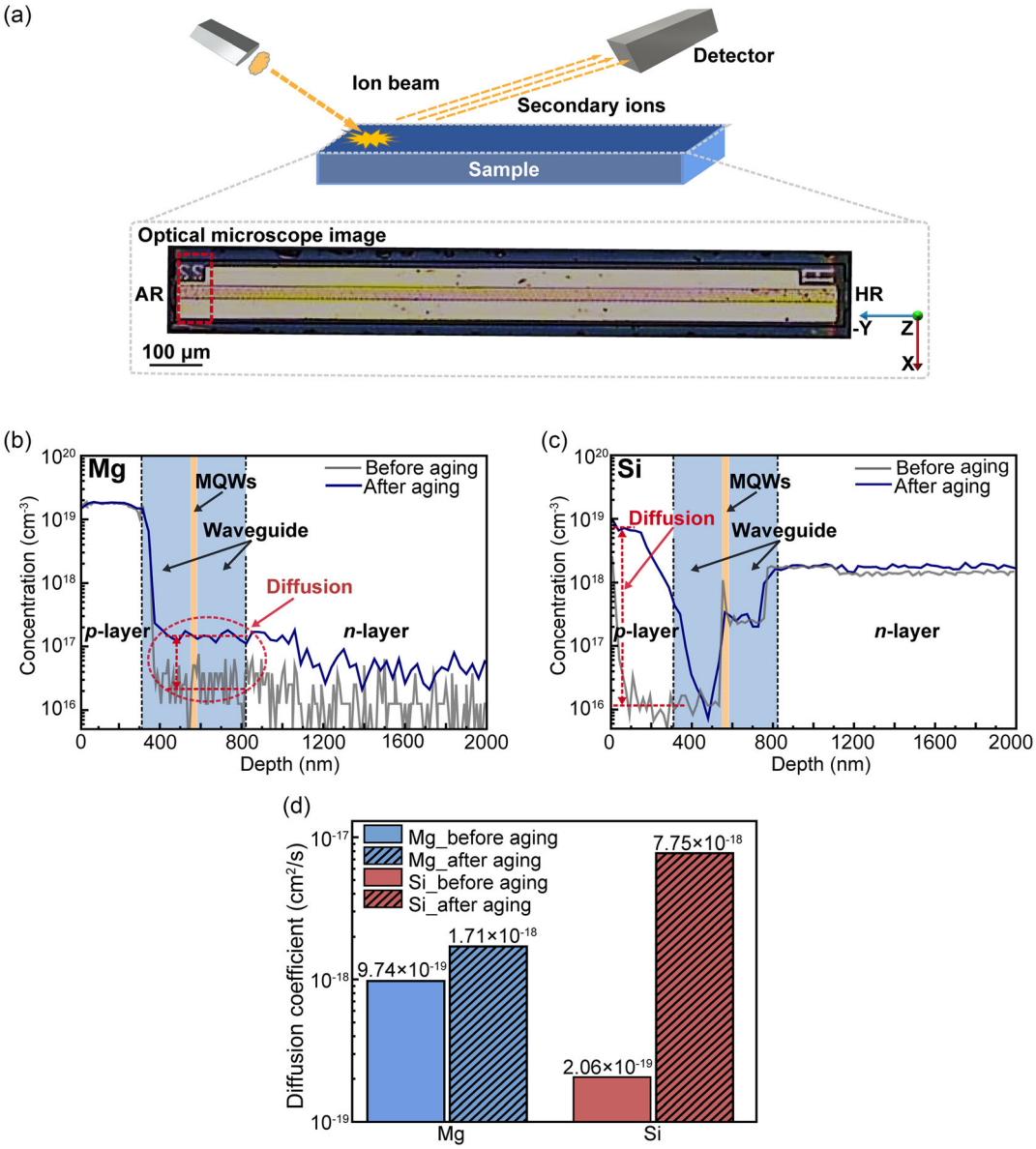

Figure 4. a) The schematic diagram of SIMS analysis. The gray dashed box shows a top view of the LD chip observed by the optical microscope, and the red dashed box indicates the analyzed area. The two ends of the LD chip were coated with an anti-reflection (AR) film and a high-reflection (HR) film, respectively. SIMS profile of b) Mg and c) Si before and after aging, analyzed from the *p*-contact layer to the *n*-cladding layer. d) Changes in the diffusion coefficients of Mg (within the active region) and Si (within the *p*-type layer) before and after aging.

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thereby increasing internal loss and accelerating LD degradation. Additionally, the diffusion of Mg and Si could induce optical transitions.[29] Specifically, Mg- and Si-related luminescence might occur in the active region under the stimulated condition. The acceptor activation energy for Mg was found to be 246 meV, and the donor activation energy for Si was 35 meV, both calculated using photoluminescence techniques.[30] Based on the activation energies of Mg and Si, the wavelengths of impurity-related luminescence within the active region were calculated to be 498 and 443 nm, respectively. These wavelengths would be selected by the resonant cavity of the LD, and the selectivity of the resonant cavity can be quantified by the quality factor (Q)[31]

$$Q = \frac{\lambda_0}{\Delta\lambda} \tag{4}$$

where λ_0 is the peak wavelength and $\Delta\lambda$ is the full-width-at-half maximum (FWHM). The Q value of the LD was calculated to be ≈492, indicating high selectivity of the resonant cavity to wavelengths close to the resonant wavelength. However, the impurity-related luminescence wavelengths (443 or 498 nm) deviated from the resonant wavelength of the LD (437 nm), causing rapid attenuation of light at these wavelengths. The impurity-related luminescence could affect the effective optical gain in the active region, resulting in decreased light output power.

2.1.2. Optical Loss Induced by Cavity Surface Degradation

In addition to the internal loss of the epilayer, mirror loss at the cavity surface is another critical component contributing to optical loss in LDs. The degradation of the cavity surface primarily stems from a high density of dangling bonds present on cleaved facets. Exposure of these dangling bonds to oxygen or photons can result in the formation of oxide components, introducing high-density defects.[32] Nonradiative recombination within these defects generates heat on the cavity surface. Accumulation of heat can damage the mirror film and lead to diffusion of mirror coating from the cavity surface. To investigate the diffusion of silicon dioxide (SiO_2) mirror coating, SIMS analysis was conducted adjacent to the anti-reflection (AR) film (excluding the AR film itself), as depicted in Figure 4a. The SIMS results revealed significant Si diffusion in the p-type layer of the LD, with a diffusion depth of nearly 500 nm (Figure 4c). The ion-beam bombardment was directed from the top of the area near the ridge region to the n-type layer, indicating that the increased Si did not diffuse from the n-type layer during SIMS analysis. This diffusion mainly originated from SiO_2 mirror film (from the cavity surface into the LD epilayer), and it could be from the SiO_2 insulation on the sidewall. High-temperature stress near the cavity surface during operation could damage the mirror film, facilitating the diffusion of mirror coating into the p-type layer. This diffusion would introduce additional defects on the cavity surface, further accelerating its degradation.

Meanwhile, the diffusion of mirror coating could degrade the effective reflection and transmission of the mirror, thereby reducing the quality and effectiveness of the mirror and increasing mirror loss. Mirror loss is determined by the reflectivity of AR and high-reflection (HR) films, expressed as[15]

$$\alpha_m = \frac{1}{2L}\ln\frac{1}{R_1 R_2} \tag{5}$$

where L is the cavity length, and R_1 and R_2 are the reflectivity of the AR and HR films, respectively. The diffusion of the mirror coating could change its reflectivity in the local area, which might negatively impact laser beam stability and lead to increased mirror loss.

Moreover, this diffusion could deteriorate the flatness of the mirror film, thereby altering local cavity length. The resonant wavelength of LDs is influenced by cavity length and can be represented as[15]

$$2n_R L = m\lambda \tag{6}$$

where n_R is the refractive index in the active region, m is an arbitrary integer, and λ is the resonant wavelength. Even subtle changes in cavity length could affect standing wave conditions in the resonant cavity. This led to fluctuations in light output power and shifts of resonant wavelength, ultimately decreasing the quality factor of the resonant cavity.

2.1.3. Deterioration of Carrier Injection

Impurity diffusion not only affects optical loss in both the epilayer and cavity surface but also impacts carrier injection. The diffusion of Mg as an acceptor toward the active region could alter the carrier distribution within the p–n junction (Figure 4b). However, the diffusion of Si in the p-type layer had a more pronounced effect, exacerbating this alteration (Figure 4c). Si elements diffusing into the p-type layer acted as donor impurities, reducing the effective carrier density by compensating for holes. P-doping in GaN is challenging due to the high ionization energy of Mg acceptors, which results in an imbalance in carrier distribution.[33] The reduction in effective hole density exacerbated this imbalance, leading to carrier injection deterioration. Consequently, fewer carriers could reach the active region to participate in radiative recombination, making it more difficult for LD to achieve population inversion. Changes in carrier density in the p/n-type layers also signify degradation of the p–n junction, which dominates the electrical characteristics of the LD. This degradation, which primarily results from Si diffusion, could be responsible for the measured variation in the forward current below the turn-on voltage (Figure 2d). To qualitatively analyze the impact of Si diffusion on the I–V characteristics of the LD, numerical simulations were conducted using PICS3D software. Si and Mg doping concentrations in the LD were adjusted based on SIMS data (increased Si doping concentration in the p-type layer but Mg doping concentration remained unchanged). Results (**Figure 5**a) showed that diffusion of Si in the p-type layer caused a variation in increased forward current (I_f) at voltages below the turn-on voltage. Consistency between simulated results and experimental measurements under low-injection current conditions further validated the effect of Si diffusion on the p–n junction electrical characteristics. Additionally, factors influencing impurity diffusion during LD operation must be considered, because of the significant correlation between diffusion processes and LD degradation.

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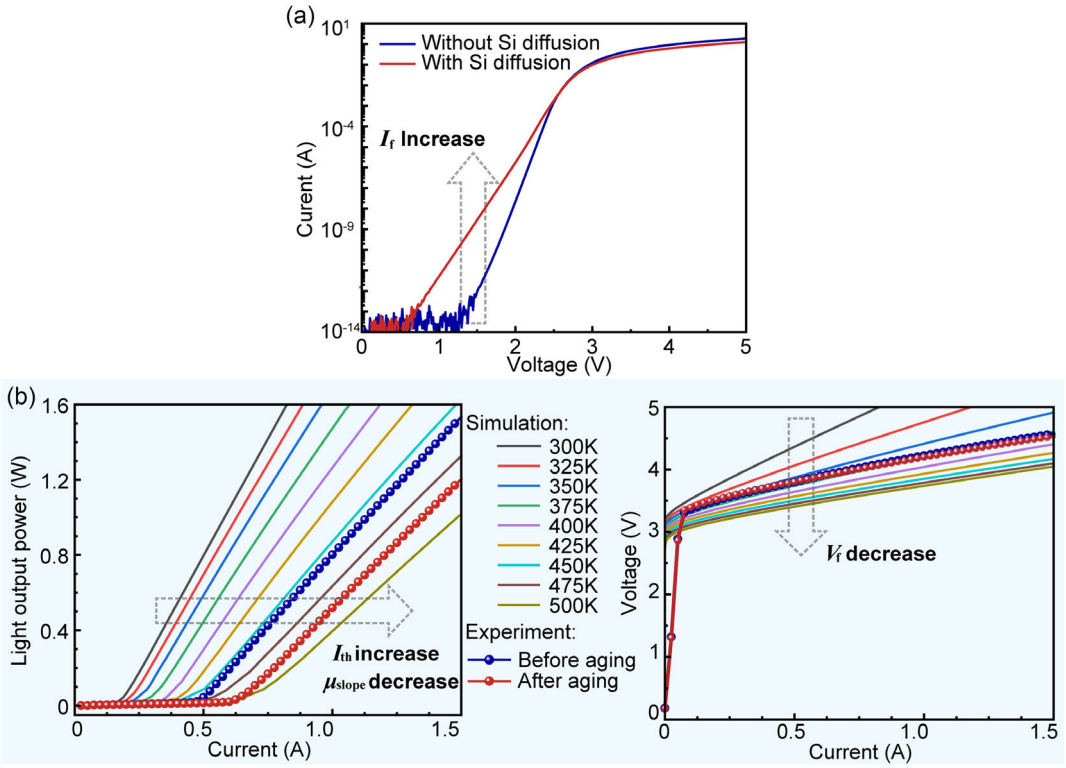

Figure 5. a) Simulated data of I–V characteristics with and without Si diffusion. b) The simulated and experimental data of L–I and I–V characteristics.

2.1.4. Temperature Effect on Impurity Diffusion and LD Performance

Temperature significantly influences impurity diffusion within LDs.[34] The diffusion process can be evaluated by changes in diffusion coefficients, which can be assessed through SIMS data. According to Fick's second law, under constant surface concentration and diffusion coefficient (D), the distribution of impurity concentration in GaN follows a complementary error function, given by[19]

$$N(x) = N_0\left(1 - \frac{2}{\sqrt{\pi}}\int_0^{\frac{x}{2\sqrt{Dt}}} e^{-x^2}\,dx\right) \tag{7}$$

where x is the diffusion depth, D is the diffusion coefficient, and t is the aging time. The values of the diffusion coefficients were obtained according to the preprocessing of SIMS data in the ref. [35]. Diffusion coefficients for Mg (within the active region) and Si (within the p-type layer) before and after aging were shown in Figure 4d. Both Mg and Si diffusion coefficients increased after aging, and the Arrhenius equation could be involved[35]

$$D = D_0 \exp\left(\frac{-E_d}{kT}\right) \tag{8}$$

where D_0 is a constant representing the diffusion coefficient at infinitely high temperatures, E_d is the diffusion activation energy, k is the Boltzmann constant, and T is the temperature. This equation elucidated the temperature sensitivity of diffusion coefficients. The notable increase in diffusion coefficients suggested increased local temperatures near the cavity surface (as the SIMS analysis area was close to the mirror film), accelerating the impurity diffusion.

The temperature rise did not occur only in the area near the cavity surface but also in the entire LD device. This temperature rise was driven by various factors, including increased optical loss and deteriorated carrier injection. These factors collectively reduced the energy conversion efficiency of the LD, resulting in more energy being converted into heat. The accumulation of heat raised the overall temperature of the device, which further exacerbated temperature-induced variations in L–I–V characteristics, such as threshold current, slope efficiency, and light output power.[36] The effects of increased junction temperature on LD performance could be investigated by numerical simulation.[37] PICS3D software was used to quantitatively analyze the impact of temperature variations on the L–I–V characteristics of the LD. The simulations covered a temperature range of 300–500 K, which encompassed the typical operating conditions of the device. The simulated L–I and I–V characteristics of the LD were illustrated together with the experimental data in Figure 5b. The simulated L–I characteristics demonstrated an increase in threshold current and a decrease in slope efficiency with rising temperature, resulting in reduced light output power. Additionally, the forward voltage (V_f) decreased as temperature increased from the simulated I–V characteristics. The variations in L–I–V characteristics could be attributed to the effect of temperature on the thermal escape of injected carriers, nonradiative recombination, optical loss, and quantum efficiency.[38] Moreover, the measured L–I curves of the sample before and

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after aging aligned with the temperature-relevant trend observed in the simulated curves (specifically in the 450–475 K and 475–500 K temperature ranges, respectively). Although the measured I–V curves showed minimal change after aging, the region below the turn-on voltage had a consistent increase in forward current in both experimental (after aging) and simulated (increasing temperature) data. These consistencies between the simulated and experimental data indicated that temperature played a crucial role in the degradation of LD performance. In addition, the L–I–V characteristics of the LD before and after aging consistently followed temperature-relevant trends within the operational temperature range (300–500 K). This indicated that effective thermal management in the high-power LD ensured its operating temperature remained within a controllable range during the aging test.

2.1.5. Relationship Between Different Degradation Factors in High-Power GaN-Based Blue LDs

The degradation of high-power blue GaN-based LDs is a multiple-factor process, and the relationship between different degradation factors reported in previous literature and investigated in this work was analyzed and illustrated (**Figure 6**). Among these factors, the degradation of the epilayer and the resonant cavity could mainly contribute to LD degradation.[12] Before the fabrication of our samples, these main factors had been considered and tried to be avoided (such as improving the epitaxy techniques, utilizing high-quality substrates, improving the deposition technique for mirror coating, etc.). Even so, the degradation phenomenon was still observed. Therefore, multiple factors contributing LD degradation (including reported and new found) were analyzed together and showed in the figure.

The structural damage and dislocation multination have been reported by the previous literature, which leads to degradation of epilayer.[6b,22,39] In the subsequent experiment, the structural damage and dislocation factors have been ruled out through TEM observation of MQWs. In addition to those two factors, the increased point defects in the active region (involving the native defects and diffused impurities) could reduce LD efficiency.[8a,19,21,26,27] The results of CL and SIMS indicated the impurities diffused into the active region increased internal loss by acting as point defects, and impurity-related luminescence further reduced the effective optical gain of the active region. Meanwhile, the degradation of the resonant cavity includes oxidation of the cavity surface and formation of carbon deposition and COD.[6a,6e,6g,6h,32] Additionally, the diffusion of SiO_2 mirror coating observed by SIMS, also indicated the degradation of the cavity surface, increasing mirror loss and reducing the quality factor of the resonant cavity. The diffusion concentration of Si into the *p*-type layer was nearly two orders of magnitude greater than that of Mg into the active region (Figure 4b,c). This means the diffusion of the mirror coating could be the key factor in LD degradation. Furthermore, interdiffusion of impurities in the *p*/*n*-type layers, such as Si diffusion into the *p*-type layer, compensated for holes and contributed to *p*–*n* junction degradation.[40] This diffusion led to the deterioration of carrier injection, making population inversion of the LD to be more

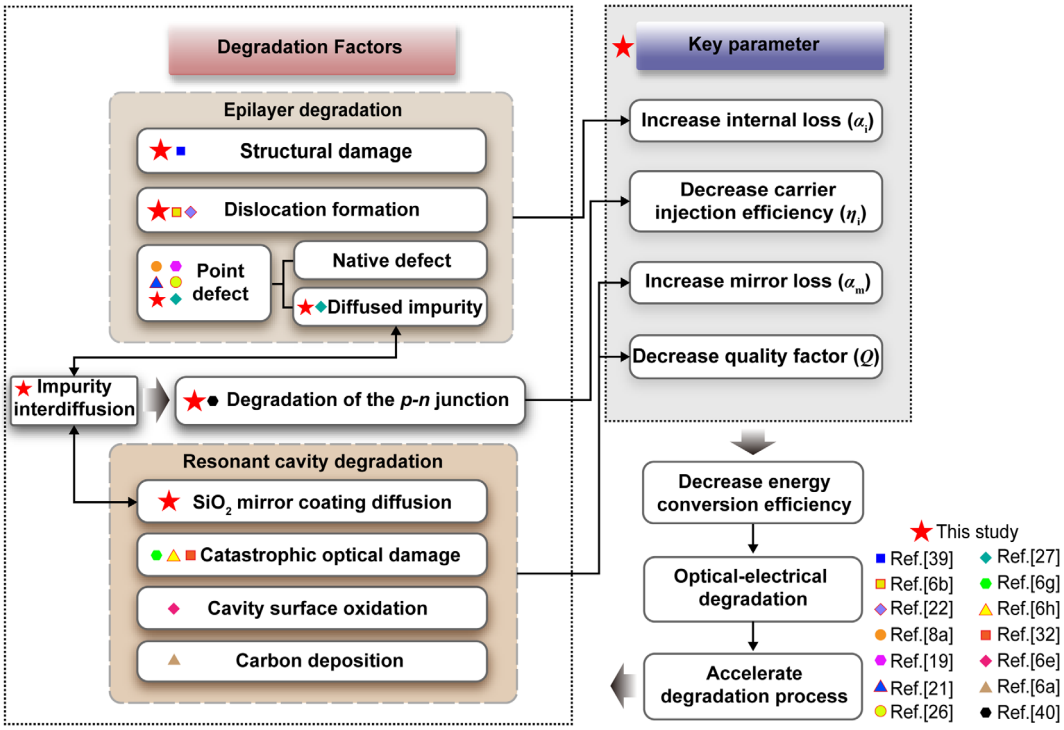

Figure 6. The multiple-factors relationship for high-power GaN-based blue LD degradation mechanism. The previously reported factors are indicated by various colored symbols, and factors analyzed in this work are marked with red stars.

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difficult. This deterioration manifested in a higher threshold current required for lasing after aging. The ratio $(\eta_{\text{i_before}} \times \alpha_{\text{m_before}})/(\eta_{\text{i_after}} \times \alpha_{\text{m_after}})$ was calculated, and the calculated value was less than 1. This suggested that the increase in mirror loss of the aged sample surpassed the decline in injection efficiency. These factors collectively decreased LD energy conversion efficiency, which resulted in localized and overall temperature rises within the LD device. This increase in temperature further accelerated LD degradation. After analyzing and checking these degradation factors (Figure S1), the diffusion of mirror coating appeared to be the most critical degradation factor. Developing new anti-aging technologies to suppress this diffusion and enhance mirror-film reliability is crucial.

2.2. Anti-aging Technology for Degradation Suppression in High-Power LDs

According to the mechanisms analyzed earlier, the diffusion of SiO_2 mirror coating could be a key factor contributing to LD degradation. To enhance the lifespan of high-power LDs, it is imperative to suppress the diffusion of the mirror coating, improving its reliability. For these purposes, an anti-aging technology was developed in this study for high-power LDs by designing and fabricating a new structure (**Figure 7**a). An AlN layer was introduced between the GaN and mirror film as the passivation layer to suppress the diffusion of the mirror coating. Pre-depositing a passivation layer before the mirror film is an effective method

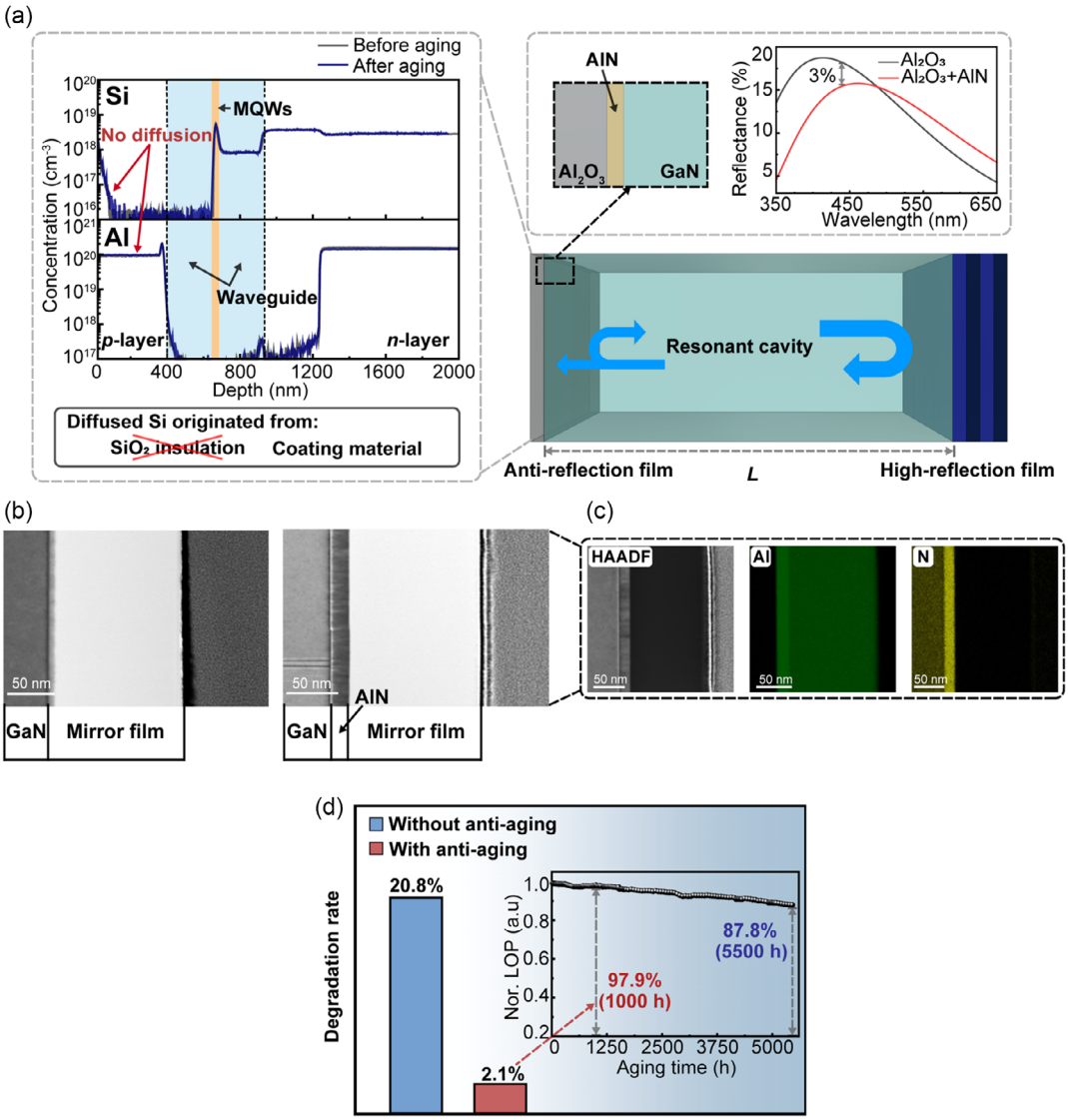

Figure 7. a) Schematic diagram of the anti-aging technology with a new structure. The structure involved the introduction of an AlN passivation layer and the replacement of mirror film. The left sub-figure was the SIMS profiles of Si and Al before and after aging. The upper right sub-figure showed the calculated reflectivity of the mirror film without and with the AlN layer. b) TEM images of the GaN–mirror film interface without and with the AlN layer. c) High-angle annular dark-field images of the AlN layer. d) The comparison of the degradation rates between the samples without and with anti-aging technology for 1000 h of aging. The sub-figure inset was the result of a 5500-h aging test for the sample with anti-aging technology and the Nor. LOP on the y-axis represented the normalized light output power.

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to protect the GaN facet. The materials used for the passivation layer must have low conductivity, stable physical properties, and must not react with GaN. AlN has a direct band gap of ≈6.1 eV, which is much broader than that of GaN (3.4 eV). This can effectively prevent the recombination of electrons and holes on the cavity surface, eliminate surface absorption, and reduce the accumulation of heat. Meanwhile, AlN has a high melting point (2473 K) and high thermal conductivity (170 W m^{-1} K^{-1}), which are much higher than those of SiO_2.[41] Additionally, AlN material does not contain oxygen, which avoids the interdiffusion of oxygen and GaN and suppresses oxidation. Therefore, utilizing AlN for the passivation layer is expected to have a good passivation effect and provide a barrier to suppress the diffusion from the cavity surface into the LD epilayer. However, the introduction of the AlN passivation layer would change the reflectivity of the mirror film. This change would affect the threshold current and the slope efficiency of the LD, further changing its light output power. Hence, the thickness of the AlN layer should be considered to minimize its impact on reflectivity. For a single layer of mirror film, the reflectivity can be calculated using the Fresnel equation, and the reflectivity of the multilayer was calculated based on the Fresnel equation and matrix algebra.[42] The calculation of mirror-film reflectivity showed that the reflectivity only changed by 3% compared to the mirror film without AlN when the thickness of AlN was set to 20 nm. This slight change indicated that introducing a 20 nm AlN layer had minimal impact on the output properties of the LD. Therefore, an AlN layer with a thickness of 20 nm was introduced between the GaN and mirror film as the passivation layer. The front facet of the sample with an AlN passivation layer was observed using TEM (Figure 7b,c), which showed a clear boundary at the GaN–mirror film interface.

In addition to the introduction of the AlN passivation layer, this anti-aging technology involved the material replacement of the mirror film. The reliability of the mirror film is closely associated with the physical properties of its coating material, such as thermal conductivity and melting point.[43] The mirror film on the facet could be easily damaged due to the high-power density and increased temperature. Aluminum oxide (Al_2O_3) has been proven to have better reliability compared to SiO_2 as the coating material for the mirror film, as demonstrated by our research group.[4] The melting point of Al_2O_3 is 2323 K (which is 673 K higher than that of SiO_2) and its thermal conductivity is 5.9 times that of SiO_2.[44] Hence, the SiO_2 mirror film was replaced with Al_2O_3 (i.e., Al_2O_3-coated samples) to improve the reliability of the mirror film. To validate the capacity of this anti-aging technology for degradation suppression, a longer aging test (5500 h) was conducted under the same conditions on the sample with this technology. A SIMS test was conducted to investigate the diffusion of the mirror coating. The SIMS result revealed that there was no diffusion of aluminum (Al) (Al_2O_3 mirror film) and Si (SiO_2 insulation material) into the *p*-type layer. This result confirmed that the diffusion of Si in the *p*-type layer of the SiO_2-coated sample originated from the mirror film rather than SiO_2 insulation. Meanwhile, the absence of the Al diffusion also indicated that this anti-aging technology effectively suppressed the diffusion of the mirror coating, ultimately improving its reliability. The aging test results showed that the normalized light output power decreased to 97.9% at 1000 h of aging, eventually declining to 87.8% at 5500 h of aging (Figure 7d). Since the 1000-h aging, the degradation suppression of the sample with anti-aging technology had improved nearly 10 times compared to the sample without this technology. This improvement in degradation suppression indicated that the new structure with the introduction of an AlN passivation layer effectively extended the LD lifespan.

3. Conclusion

This study demonstrated the degradation mechanisms of high-power GaN-based LDs. Degradation analysis used L–I–V characterization, TEM, CL, and SIMS. Factors that influenced the LD degradation (including impurity interdiffusion within the LD epilayer) were analyzed. Encouragingly, the diffusion of mirror coating was the first time to be found as a key factor for LD degradation, which had not been reported in milliwatt-level LDs. Specifically, the diffusion of SiO_2 mirror coating from the cavity surface into the *p*-type layer emerged, indicating the degradation of the resonant cavity surface. Numerical simulations conducted via PICS3D further elucidated the effects of elemental diffusion and temperature increases on LD degradation. In response to these findings, an anti-aging technology was developed through the design and fabrication of a new structure, aiming to suppress diffusion and enhance the reliability of the mirror film. This technology introduced an AlN passivation layer at the GaN–mirror film interface to serve as a barrier against diffusion and replaced the SiO_2 mirror coating with Al_2O_3. The efficacy of the anti-aging technology was validated, and a nearly ten-time improvement in degradation suppression was achieved since 1000-h aging. This study not only elucidated the degradation mechanisms of high-power GaN-based LDs but also developed a technique that significantly extended LD lifespan, thereby advancing their practical applications.

4. Experimental Section

Fabrication: The high-power GaN-based blue LD was fabricated by metal-organic chemical vapor deposition (MOCVD) on a c-plane free-standing GaN substrate (dislocation density is less than 3×10^6 cm^{-2}). The epitaxial structure included an *n*-GaN layer, *n*-type cladding layer, *n*-type waveguide layer, asymmetric InGaN MQWs, undoped waveguide layer, *p*-type electron-blocking layer (EBL), superlattice (SL) cladding layer (with 60 periods), and *p*-type contact layer. Subsequently, the chips further went through the standard process flow of edge-emitting LDs. The LD device was fabricated in a ridge-waveguide structure with a ridge width of 30 μm and a cavity length of 1200 μm. In order to form the *p*-type contact, the indium tin oxide (ITO) layer was deposited with a thickness of 200 nm and annealed at 575 °C. SiO_2 (300 nm) was utilized for insulation to protect the sidewall and ensure that the current could be injected from the ridge. Chromium/platinum/gold and titanium/platinum/gold were applied as the *p*-electrode pad and *n*-electrode pad, respectively. A single SiO_2 layer was deposited for AR film, and the HR film was prepared by coating the back cavity surface with a 1/4 alternating layer composed of titanium dioxide (TiO_2) and SiO_2. The reflectivity of the AR films and HR films were about 18% and 99%, respectively. In terms of the packaging process, LDs were flip-mounted and packaged into the TO9 package (diameter of 9 mm).

Characterizations: The L–I characteristics of the aged samples were measured using a semi-automatic chip probe station (MPI, LBP-100) at various periods during the aging test. The I–V characteristics of the samples were measured using a continuous-wave power supply

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(Tektronix, Keithley 2400) before and after the aging test. The structural and optical characterizations were conducted in both the ridge and non-ridge regions of the aged sample. The ridge region underwent the current injection, but the non-ridge region did not. The structure of the active region was observed by the TEM (Thermo Fisher Scientific, Talos F200X). The CL mode of a scanning electron microscope (SEM) was used to analyze the optical property near the active region (including the waveguide layer). The thermal field emitter was used as the source of SEM (Orsay, ECLIPSE). The CL measurement was taken at 460 nm and conducted with an electron beam energy of 15 keV and a beam current of 3.4 nA. SIMS (Cameca, IMS 7f-Auto) was utilized to analyze the elemental distribution of impurities within samples before and after the aging test.

Numerical Simulations: The PICS3D commercial software (Crosslight Software Inc.) was used to investigate the L–I–V characteristics of the LD. The simulation structure was designed based on the epitaxial structure. In the simulations, the Auger recombination coefficient was set as 1×10^{-31} cm^{-6} s^{-1},[45] and the Shockley–Read–Hall recombination time was 1.5 ns.[46] This was because the surface charge could be shielded due to the presence of defects and free carriers, and the experimentally measured surface charge was smaller than the theoretical value.[47] Therefore, the screening factor was set to 30%,[48] and the band offset ratio ($\Delta E_c/\Delta E_g$) of the LD was set to 0.65.[49] The simulated temperature of the active region was set to 460 K. In addition, other parameters in the numerical simulation can be consulted in the ref. [50].

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

E.Z. and Y.Z. contributed equally to this study as first authors. This work was supported by the university-enterprise cooperation project of Xiamen University (grant no. 20213160A0754).

Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Enming Zhang: Data curation (equal); Formal analysis (equal); Investigation (equal); Software (lead); Visualization (equal); Writing—original draft (lead); Writing—review and editing (equal). **Yue Zeng**: Conceptualization (supporting); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Validation (equal); Writing—review and editing (equal). **Wenyu Kang**: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (supporting); Resources (supporting); Software (supporting); Supervision (equal); Validation (equal); Visualization (equal); Writing—original draft (equal); Writing—review and editing (lead). **Zhibai Zhong**: Data curation (equal); Formal analysis (equal); Investigation (equal); Software (supporting). **Yushou Wang**: Formal analysis (equal); Methodology (equal). **Tongwei Yan**: Formal analysis (supporting); Investigation (supporting); Methodology (equal); Project administration (supporting). **Shaohua Huang**: Conceptualization (equal); Formal analysis (supporting); Funding acquisition (equal); Methodology (supporting); Supervision (supporting). **Zhongying Zhang**: Conceptualization (equal); Funding acquisition (equal); Investigation (supporting); Project administration (supporting); Resources (equal); Supervision (equal). **Kechuang Lin**: Conceptualization (equal); Funding acquisition (lead); Resources (lead); Supervision (lead). **Junyong Kang**: Conceptualization (lead); Data curation (lead); Formal analysis (equal); Funding acquisition (lead); Investigation (equal); Methodology (equal); Project administration (lead); Resources (lead); Software (equal); Supervision (lead); Validation (lead); Visualization (supporting); Writing—original draft (supporting); Writing—review and editing (lead).

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

III-V semiconductors, facet passivation, impurity diffusion, PICS3D, SIMS

Received: July 15, 2024
Revised: August 19, 2024
Published online: September 4, 2024

[1] a) S. Nakamura, M. Senoh, S. Nagahama, N. Iwasa, T. Yamada, T. Matsushita, K. Hiroyuki, S. Yasunobu, *Jpn. J. Appl. Phys.* **1996**, *35*, L74; b) Z. Zhang, M. Kushimoto, T. Sakai, N. Sugiyama, L. J. Schowalter, C. Sasaoka, H. Amano, *Appl. Phys. Express* **2019**, *12*, 124003; c) S. Lutgen, D. Dini, I. Pietzonka, S. Tautz, A. Breidenassel, A. Lell, A. Avramescu, C. Eichler, T. Lermer, J. Müller, G. Bruederl, A. Gomez-Iglesias, U. Strauss, W. Scheibenzuber, U. Schwarz, B. Pasenow, S. Koch, in *Novel In-Plane Semiconductor Lasers X* (Eds: A. Belyanin, P. Smowton), SPIE, California **2011**, p. 79530G. d) S. Gerhard, L. Nähle, B. Jentzsch, H. König, E. Reiger, U. Heine, S. Tautz, C. Eichler, G. Brüderl, T. Wurm, D. Borovac, M. Behringer, L. Kreiner, M. Jama, N. von Malm, A. Balck, M. Baumann, V. Krause, in *Gallium Nitride Materials and Devices XVIII* (Eds: H. Fujioka, H. Morkoç, T. Schwarz), SPIE, California **2023**, p. 124210G; e) H. König, M. Ali, W. Bergbauer, J. Brückner, G. Brüderl, C. Eichler, S. Gerhard, U. Heine, A. Lell, L. Naehle, M. Peter, J. Ristic, G. Rossbach, A. Somers, B. Stojetz, S. Tautz, J. Wagner, T. Wurm, U. Strauss, M. Baumann, A. Balck, V. Krause, in *Novel In-Plane Semiconductor Lasers XVIII* (Eds: A. Belyanin, P. Smowton), SPIE, California **2019**, p. 10939C. f) D. Brauda, D. Krug, M. von Sivers, H. Thiem, M. Schütz, A. Bachmann, M. Malach, B. Globisch, in*Conf. Comp. Packaging Laser Syst. X* (Eds: A. Glebov, P. Leisher), SPIE, California **2024**, p. 128660D.

[2] U. Strauss, A. Somers, U. Heine, T. Wurm, M. Peter, C. Eichler, S. Gerhard, G. Bruederl, S. Tautz, B. Stojetz, A. Loeffler, H. Koenig, in *Novel In-Plane Semiconductor Lasers XVI* (Eds: A. Belyanin, P. Smowton), SPIE, California **2017**, p. 101230A.

[3] F. Liang, D. Zhao, Z. Liu, P. Chen, J. Yang, L. Duan, Y. Shi, H. Wang, *J. Semicond.* **2021**, *42*, 112801.

[4] Z. Zhong, S. Lu, J. Li, W. Lin, K. Huang, S. Li, D. Cai, J. Kang, *Opt. Laser Technol.* **2021**, *139*, 106985.

[5] M. Nakazumi, A. Oguri, Y. Ishige, R. Konishi, T. Hirao, T. Takeshi, N. Matsumoto, N. Hayamizu, H. Mori, M. Kaneko, N. Tsukiji, in *High-Power Diode Laser Technology XX* (Eds: M. Zediker, E. Zucke), SPIE, California **2022**, p. 119830J.

[6] a) C. Kim, Y. Choi, Y. H. Jang, M. K. Kang, M. Joo, M. S. Noh, in *Gallium Nitride Materials and Devices III* (Eds: M. Hadis, C. Litton, J. Chyi, Y. Nanishi, Y. Euijoon), SPIE, California **2008**, p. 68940O; b) S. Tomiya, O. Goto, M. Ikeda, *Proc. IEEE* **2010**, *98*, 1208; c) M. Marioli, M. Meneghini, F. Rossi, G. Salviati, C. De Santi, G. Mura, G. Meneghesso, E. Zanoni, *Phys. Status Solidi A* **2015**,

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212, 974; d) J. Smalc-Koziorowska, E. Grzanka, A. Lachowski, R. Hrytsak, M. Grabowski, S. Grzanka, S. Kret, R. Czernecki, H. Turski, L. Marona, T. Markurt, T. Schulz, M. Albrecht, M. Leszczynski, *ACS Appl. Mater. Interfaces* **2021**, *13*, 7476; e) X. Wang, Z. Liu, D. Zhao, P. Chen, F. Liang, J. Yang, *J. Appl. Phys.* **2021**, *129*, 223106; f) C. De Santi, M. Meneghini, S. Carraro, S. Vaccari, N. Trivellin, S. Marconi, M. Marioli, G. Meneghesso, E. Zanoni, *Microelectron. Reliab.* **2013**, *53*, 1534. g) M. Hempel, J. W. Tomm, B. Stojetz, H. König, U. Strauss, T. Elsaesser, *Semicond. Sci. Technol.* **2015**, *30*, 072001. h) J. Tomm, R. Kernke, A. Löffler, B. Stojetz, A. Lell, H. König, in*Conf. Novel In-Plane Semicond. Lasers XVII*, (Eds: A. Belyanin, P. Smowton), California **2018**, 1055308; i) C. Zhang, W. Liao, S. Yang, H. Wang, Z. Wei, G. Lu, in*2021 Int. Conf. Syst. Reliab. Saf. Eng.*, IEEE, Harbin **2021**, 67. j) T. Chuang, H. Lin, H. Wang, C. Chuang, C. Tsai, *J. Electron. Mater.* **2015**, *44*, 623.

[7] N. Trivellin, M. Meneghini, E. Zanoni, K. Orita, M. Yuri, T. Tanaka, D. Ueda, G. Meneghesso, in **2010** *IEEE Inter. Reliability Physics Symp.*, IEEE, California **2010**, 1.

[8] a) M. Meneghini, C. De Santi, N. Trivellin, K. Orita, S. Takigawa, T. Tanaka, D. Ueda, G. Meneghesso, E. Zanoni, *Appl. Phys. Lett.* **2011**, *99*, 093506; b) S. Tomiya, T. Hino, S. Goto, M. Takeya, M. Ikeda, *IEEE J. Sel. Top. Quantum Electron.* **2004**, *10*, 1277; c) P. Wen, S. Zhang, D. Li, J. Liu, L. Zhang, K. Zhou, M. X. Feng, A. Tian, F. Zhang, X. Gao, C. Zeng, H. Yang, *J. Phys. D: Appl. Phys.* **2015**, *48*, 415101.

[9] T. Kamikawa, Y. Kawaguchi, P. O. Vaccaro, S. Ito, H. Kawanishi, *Appl. Phys. Lett.* **2009**, *95*, 031106.

[10] A. Bojarska, G. Muzioł, C. Skierbiszewski, E. Grzanka, P. Wiśniewski, I. Makarowa, R. Czernecki, T. Suski, P. Perlin, *Appl. Phys. Express* **2017**, *10*, 091001.

[11] Y. Lan, G. Yang, Y. Zhao, Y. Liu, A. Demir, *Appl. Surf. Sci.* **2022**, *596*, 153506.

[12] Y. Song, Z. Lv, J. Bai, S. Niu, Z. Wu, L. Qin, Y. Chen, L. Liang, Y. Lei, P. Jia, X. Shan, L. Wang, *Crystals* **2022**, *12*, 12765.

[13] Y. Huang, J. Yang, Z. Liu, F. Liang, D. Zhao, *J. Appl. Phys.* **2023**, *134*, 095701.

[14] P. Wen, S. Zhang, J. Liu, D. Li, L. Zhang, K. Zhou, X. Su, A. Tian, F. Zhang, H. Yang, *IEEE Trans. Device Mater. Reliab.* **2016**, *16*, 638.

[15] L. Coldren, S. Corzine, M. Mašanović, in *Diode Lasers and Photonic Integrated Circuits*, John Wiley & Sons, New York **2012**.

[16] a) M. Kawaguchi, O. Imafuji, S. Nozaki, H. Hagino, S. Takigawa, T. Katayama, T. Tanaka, in *Gallium Nitride Materials and Devices X* (Eds: J. Chyi, H. Fujioka, M. Hadis), SPIE, California **2016**, 974818; b) M. Gao, J. Yang, W. Jia, D. Zhao, G. Zhai, H. Dong, B. Xu, *Nanomaterials* **2024**, *14*, 649.

[17] N. Tessler, G. Eisenstein, *IEEE Photonics Technol. Lett.* **1993**, *5*, 291.

[18] S. Chao, W. Wang, C. Lee, *Appl. Opt.* **2001**, *40*, 2177.

[19] K. Orita, M. Meneghini, H. Ohno, N. Trivellin, N. Ikedo, S. Takigawa, M. Yuri, T. Tanaka, E. Zanoni, G. Meneghesso, *IEEE J. Quantum Electron.* **2012**, *48*, 1169.

[20] C. De Santi, M. Meneghini, M. Marioli, M. Buffolo, N. Trivellin, T. Weig, K. Holc, K. Köhler, J. Wagner, U. T. Schwarz, G. Meneghesso, E. Zanoni, *Microelectron. Reliab.* **2014**, *54*, 2147.

[21] D. Monti, M. Meneghini, C. De Santi, G. Meneghesso, E. Zanoni, A. Bojarska, P. Perlin, *Microelectron. Reliab.* **2017**, *76*, 584.

[22] T. Weatherley, W. Liu, V. Osokin, D. Alexander, R. Taylor, J. Carlin, R. Butté, N. Grandjean, *Nano Lett.* **2021**, *21*, 5217.

[23] M. La Grassa, M. Meneghini, C. De Santi, E. Zanoni, G. Meneghesso, *Microelectron. Reliab.* **2016**, *64*, 614.

[24] M. Meneghini, G. Meneghesso, N. Trivellin, E. Zanoni, K. Orita, M. Yuri, D. Ueda, *IEEE Electron Device Lett.* **2008**, *29*, 578.

[25] a) J. Neugebauer, C. Van de Walle, *Phys. Rev. B* **1994**, *50*, 8067; b) S. Limpijumnong, C. Van de Walle, *Phys. Rev. B* **2004**, *69*, 035207; c) M. A. Reshchikov, H. Morkoç, *J. Appl. Phys.* **2005**, *97*, 061301.

[26] M. Rossetti, T. Smeeton, W. Tan, M. Kauer, S. Hooper, J. Heffernan, H. Xiu, C. J. Humphreys, *Appl. Phys. Lett.* **2008**, *92*, 151110.

[27] O. Nam, K. Ha, J. Kwak, S. Lee, K. Choi, T. Chang, S. Chae, W. Lee, Y. Sung, H. Paek, J. Chae, T. Sakong, J. Son, H. Ryu, Y. Kim, Y. Park, *Phys. Status Solidi A* **2004**, *201*, 2717.

[28] L. Marona, P. Perlin, R. Czernecki, M. Leszczynski, M. Bockowski, R. Jakiela, T. Suski, S. P. Najda, *Appl. Phys. Lett.* **2011**, *98*, 241115.

[29] F. Zimmermann, J. Beyer, C. Röder, F. Beyer, E. Richter, K. Irmscher, J. Heitmann, *Phys. Status Solidi A* **2021**, *218*, 2100235.

[30] K. Subba Ramaiah, Y. Su, S. Chang, F. Juang, C. Chen, *J. Cryst. Growth* **2000**, *220*, 405.

[31] M. Hwang, H. Lee, K. Kim, K. Jeong, S. Kwon, K. Koshelev, Y. Kivshar, H. Park, *Nat. Commun.* **2021**, *12*, 4135.

[32] C. Silfvenius, P. Blixt, C. Lindstrom, A. O. Feitisch, in *High-Power Diode Laser Technology and Applications II* (Eds: M. Zediker), SPIE, California **2004**, p. 132.

[33] T. Narita, H. Yoshida, K. Tomita, K. Kataoka, H. Sakurai, M. Horita, M. Bockowski, N. Ikarashi, J. Suda, T. Kachi, Y. Tokuda, *J. Appl. Phys.* **2020**, *128*, 090901.

[34] a) Y. Itoh, S. Lu, H. Watanabe, M. Deki, S. Nitta, Y. Honda, A. Tanaka, H. Amano, *Appl. Phys. Express* **2022**, *15*, 116505; b) C. De Santi, M. Meneghini, G. Meneghesso, E. Zanoni, *Microelectron. Reliab.* **2016**, *64*, 623.

[35] M. Pagani, *J. Appl. Phys.* **1990**, *68*, 3726.

[36] Z. Zhang, M. Kushimoto, A. Yoshikawa, K. Aoto, C. Sasaoka, L. Schowalter, H. Amano, *Appl. Phys. Lett.* **2022**, *121*, 222103.

[37] a) K. Mehta, Y. Liu, J. Wang, H. Jeong, T. Detchprohm, Y. Park, S. R. Alugubelli, S. Wang, F. Ponce, S. Shen, R. Dupuis, P. Yoder, *J. Quantum Electron.* **2018**, *54*, 1; b) B. Lee, H. Ryu, *Opt. Quantum Electron.* **2023**, *55*, 154.

[38] P. Epperlein, in *Semiconductor Laser Engineering, Reliability and Diagnostics*, John Wiley & Sons, New York **2013**.

[39] P. Wen, S. Zhang, D. Li, J. Liu, L. Zhang, D. Shi, K. Zhou, A. Tian, S. Feng, H. Yang, *Superlattices Microstruct.* **2016**, *99*, 72.

[40] H. Xiu, E. Thrush, L. Zhao, A. Phillips, C. Humphreys, in *2017 China Int. Forum Solid State Light. Int. Forum Wide Bandgap Semicond. China*, IEEE, Beijing **2017**, p. 38.

[41] a) E. Schwarzer-Fischer, U. Scheithauer, A. Michaelis, *Ceramics* **2023**, *6*, 416; b) R. Xu, M. Muñoz Rojo, S. Islam, A. Sood, B. Vareskic, A. Katre, N. Mingo, K. Goodson, H. Xing, D. Jena, E. Pop, *J. Appl. Phys.* **2019**, *126*, 185105.

[42] S. Lian, Y. Yu, S. Lin, G. Lin, C. Xu, J. Wang, *Mater. Sci.* **2017**, *07*, 78.

[43] J. Souto, J. Pura, J. Jiménez, *J. Phys. D: Appl. Phys.* **2019**, *52*, 343002.

[44] G. Chen, M. Craven, A. Kim, A. Munkholm, S. Watanabe, M. Camras, W. Götz, F. Steranka, *Phys. Status Solidi A* **2008**, *205*, 1086.

[45] M. Brendel, A. Kruse, H. Jönen, L. Hoffmann, H. Bremers, U. Rossow, A. Hangleiter, *Appl. Phys. Lett.* **2011**, *99*, 031106.

[46] K. Baik, Y. Irokawa, F. Ren, S. Pearton, S. Park, Y. Park, *Solid-State Electron.* **2003**, *47*, 1533.

[47] V. Fiorentini, F. Bernardini, O. Ambacher, *Appl. Phys. Lett.* **2002**, *80*, 1204.

[48] X. Li, D. Zhao, D. Jiang, P. Chen, Z. Liu, J. Zhu, J. Yang, W. Liu, X. He, X. Li, F. Liang, L. Zhang, J. Liu, H. Yang, *Phys. Status Solidi A* **2016**, *213*, 2223.

[49] I. Vurgaftman, J. Meyer, L. Ram-Mohan, *J. Appl. Phys.* **2001**, *89*, 5815.

[50] F. Liang, D. Zhao, D. Jiang, Z. Liu, J. Zhu, P. Chen, J. Yang, W. Liu, S. Liu, Y. Xing, *Jpn. J. Appl. Phys.* **2018**, *57*, 070307.

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